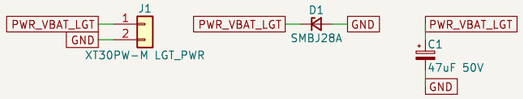
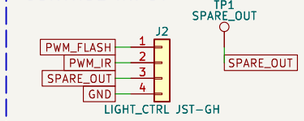
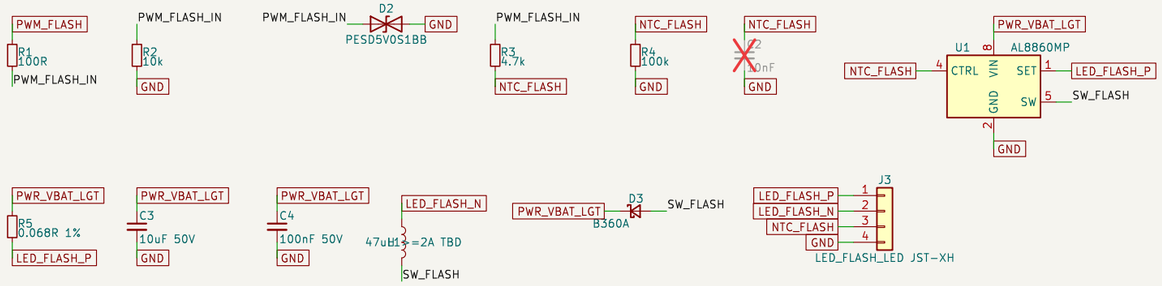
## POWER INPUT



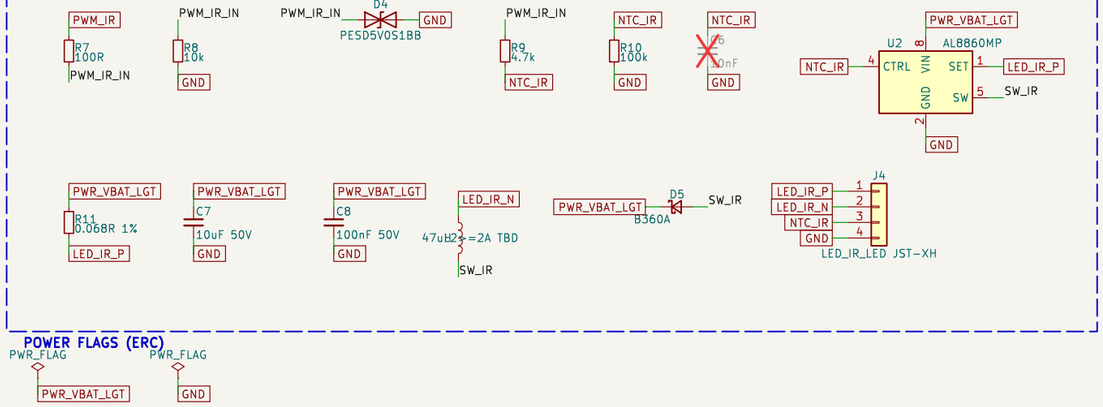

## CONTROL INPUT




## FLASHLIGHT CHANNEL (CH1)




## IR CHANNEL (CH2)




## MECHANICAL




## POWER FLAGS (ERC)




## NOTES

1. LED current = 0.1 V / R_SENSE (R5, R11): 0.068R is about 1.5 A. Set to the emitter's rating and verify the AL8860 sense voltage.
2. CTRL network: R3/R9 4.7k from the PWM input, R4/R10 100k to GND, NTC (10k B3950) from CTRL to GND at the emitter. Hot NTC pulls CTRL down and reduces current.
3. Check the AL8860 CTRL voltage range and internal pull-up, then adjust R3/R4 (R9/R10).
4. C2/C6 (DNP) turn high-frequency PWM into smooth analog dimming when fitted.
5. On 4S packs the LED string's forward voltage must stay below about 11 V.

Rev A bootstrap
Bootstrapped by hardware/tools/gen_kicad.py; edit in KiCad from here on
PWM dimming via CTRL; NTC on the emitter heatsink derates current
Two AL8860 constant-current LED drivers: flashlight (CH1) and IR (CH2)
**Jetson drone electronics**

Sheet: /
File: lgt.kicad_sch

# Title: LGT — Lighting Driver Board

| | | |
|---|---|---|
| Size: A3 | Date: 2026-11-06 | **Rev: A** |
| KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 | | Id: 1/1 |